

# A barrier for vector-valued smoothing certificates for Sidon sets

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## Abstract

Let  $h(N)$  be the maximum size of a Sidon set contained in  $\{1, \dots, N\}$ . Successive refinements of the Erdős–Turán argument have reduced the constant  $\gamma$  in  $h(N) \leq N^{1/2} + \gamma \cdot N^{1/4} + O(1)$  from 1 to 0.9435, the most recent improvements being produced by numerical certificates inside the vector-valued smoothing framework of Hou and Zhao. We study the infimum  $\gamma^*$  of the constants certifiable in that framework and in a two-sided extension of it, admitting asymmetric kernels, which the same proof covers. We prove that  $\gamma^* \geq \sqrt{((\pi^2+16)/32)} = 0.899124\dots$  in both settings, using an explicit dual multiplier together with a sharp inequality  $\|\rho\|_{2^2} \cdot \mathbb{E}|X-Y| \geq \pi^2/32$  whose extremal densities are cosine arches. In the other direction we identify the optimal dual multiplier as twice the renewal measure of the uniform distribution on  $[0,1]$ , and derive from it an explicit certificate — the linear kernel  $\rho(x) = 2x$ , with boundary weights  $e^t - 1$  at one end and 1 at the other — that attains  $\gamma = 2\sqrt{2}/3 = 0.942809\dots$  in the continuum limit. Its discretization at resolution  $m = 512$ , verified in exact rational arithmetic, gives  $\gamma^* < 0.942809522$ . We conjecture that  $\gamma^* = 2\sqrt{2}/3$ . Either way this family of arguments is bounded away from the error term conjectured in Erdős Problem 30 by a proved constant, and if the conjecture holds the remaining room inside it is below  $5 \times 10^{-7}$ .

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## 1. Introduction

A set of integers is a *Sidon set* (a  $B_2$  set) if all of its pairwise sums  $a + b$  with  $a \leq b$  are distinct, equivalently if its non-zero differences are distinct. Write

$$h(N) = \max\{ |A| : A \subseteq \{1, \dots, N\}, A \text{ a Sidon set} \}.$$

Erdős and Turán [5] proved  $h(N) \leq N^{1/2} + O(N^{1/4})$ ; their argument, in the form given by Lindström [8], yields  $h(N) \leq N^{1/2} + N^{1/4} + 1$ . Erdős Problem 30 [2, 9] asks whether

$$h(N) = N^{1/2} + o_\varepsilon(N^\varepsilon) \quad \text{for every } \varepsilon > 0, \quad (1.1)$$

which remains open. In eighty-five years the exponent  $1/4$  in the error term has never been improved. What has moved is the constant in front of it: 0.998 by Balogh, Füredi and Roy [1], 0.99703 by O’Bryant [10], 0.98183 by Carter, Hunter and O’Bryant [3], 0.97633 announced by Carter, Georgiev, Gómez-Serrano, Hunter, O’Bryant, Tao and Wagner, and 0.9435 in the current version of Hou and Zhao [7]. The last of these is of a different character from its predecessors. Hou and Zhao recast the whole family of

smoothing arguments as a finite optimization problem, in which a bound is certified by a vector of kernels, a vector of boundary weights and a set of mixing weights satisfying an explicit system of linear inequalities; improvements are then obtained by running an optimizer over that system.

This paper asks how far that road goes. We work with a slight extension of the framework, in which the kernels need not be symmetric and each end of the interval carries its own boundary weights; Proposition 2.4 shows that the proof in [7] covers it after a change to one step. Write  $\gamma^*_{\text{sym}}$  for the infimum of the constants certifiable in the Hou-Zhao framework (Definition 2.2) and  $\gamma^*$  for the infimum over its two-sided extension, so that  $\gamma^* \leq \gamma^*_{\text{sym}}$ . Every bound obtainable by optimizing inside either framework — with more kernels, finer grids or deeper boundary layers — has constant at least  $\gamma$ . *Our results bracket  $\gamma$  from both sides.*

**Theorem A.**  $\gamma^*_{\text{sym}} \geq \gamma^* \geq \sqrt{(\pi^2 + 16)/32} = 0.89912465\dots$

**Theorem B** (computational).  $\gamma^* < 0.942809522$ . The bound is attained by an explicit two-sided certificate at resolution  $m = 512$ ,  $L = 8$ , with the single linear kernel  $\rho(x) = 2x$ , whose 8194 covering inequalities and whose constants  $a$ ,  $b$  and  $\gamma^2$  have been verified in exact rational arithmetic. In the continuum relaxation the same kernel attains  $a \cdot b = 8/9$  exactly (Theorem 5.3), and its discretizations approach  $\gamma = 2\sqrt{2}/3$  with  $a \cdot b - 8/9 \approx 2/(9m^2)$ .

Together these give

$$0.899124 \leq \gamma^* < 0.9428096, \quad (1.2)$$

with  $2\sqrt{2}/3 = 0.94280904\dots$  inside, approached from above by the explicit certificate of Section 5 and not undercut by any numerical optimization we have run. The two sides are independent: an error in the optimizer would not touch Theorem A, and an error in Theorem A would not touch the verified certificates of Theorem B. In particular the framework cannot certify an error term of size  $o(N^{1/4})$ , let alone the  $N^\epsilon$  of (1.1): the lineage of arguments it generalizes is bounded away from the conjecture by a proved constant, not merely by a computed one.

The upper side of (1.2) came from reading the optimal dual multiplier off the numerics. It is twice the renewal measure of the uniform distribution on  $[0,1]$ ; feeding that measure back into the optimality conditions determines the kernel's autocorrelation, and hence the kernel: the linear density  $2x$ , which is asymmetric. Symmetric certificates can only approach it through mixtures, which is why a two-sided certificate at resolution  $m$  does about as well as a symmetric one at  $2m$  (Section 4.6). We conjecture that  $\gamma^* = 2\sqrt{2}/3$  exactly.

The proof of Theorem A is short and uses no computation. For fixed kernels, the optimization over boundary weights is a convex quadratic program, so any nonnegative multiplier applied to the covering constraint yields a valid lower bound by weak duality. Taking the multiplier to be  $4 \cdot (\text{Lebesgue measure on } [0,\infty))$  gives  $\gamma^* \geq \pi/(4\sqrt{2}) = 0.555360\dots$ . Adding a single atom at the origin, with the correct weight, contributes a term that is *independent of the kernel*, because for any density  $\rho$  with distribution function  $R$ ,

$$\int_0^\infty (1 - R(t)) \rho(t) dt = \int_0^1 (1 - R) dR = 1/2. \quad (1.3)$$

What is left is the scale-invariant functional  $Q(\rho) = \|\rho\|_{2^2}^2 \cdot E|X - Y|$ , whose sharp lower bound  $\pi^2/32$ , attained by a cosine arch, is proved in Theorem 3.3 by a quantile substitution and one application of Cauchy-Schwarz. The two pieces combine to  $a \cdot b \geq \pi^2/32 + 1/2$ , which is Theorem A.

**Conditionality.** The implication from a certificate to a bound on  $h(N)$  is Lemma 2.1 of [7], restated as Proposition 2.1 below; that preprint has not yet been refereed, and every statement here about  $h(N)$  inherits that dependence. Theorems A and B do not: they are statements about a finite optimization problem and its continuum limit, and hold whatever the status of Proposition 2.1.

Section 2 sets up the framework and its two-sided extension, and shows that every discrete certificate is a certificate for its continuum relaxation with the same constants, so that a continuum bound applies at every resolution without a limiting argument. Section 3 proves Theorem A. Section 4 reports the numerical scans, the effect of asymmetry and the structure of the limiting certificates. Section 5 identifies the optimal dual multiplier, derives from it the certificate of Theorem 5.3, and collects consequences and open questions.

## 2. The certificate framework

### 2.1 Discrete certificates

Fix integers  $m \geq 1$  (bins per kernel),  $L \geq 1$  (boundary blocks) and  $K \geq 1$  (kernels). A *certificate* at resolution  $(m, L, K)$  consists of

- kernels  $p^\wedge(k) \in \mathbb{R}^m$ ,  $k = 1, \dots, K$ , nonnegative and symmetric, with  $\sum_i p^\wedge(k)_i = 1$ ;
- boundary weights  $w^\wedge(k) \in \mathbb{R}^{(Lm)}$ , extended by  $w^\wedge(k)_j = 1$  for  $j \geq Lm$ ;
- mixing weights  $\lambda_k \geq 0$  with  $\sum_k \lambda_k = 1$ ,

subject to the covering inequalities

$$\sum_k \lambda_k \sum_{i=0}^{m-1} p^\wedge(k)_i w^\wedge(k)_{\{q+i\}} \geq 1 \quad (q = 0, 1, \dots, Lm). \quad (2.1)$$

Associated with a certificate are the two constants

$$a = m \cdot \sum_k \lambda_k \sum_i (p^\wedge(k)_i)^2, \quad b = 1 + 2 \left( \sum_k \lambda_k \sum_{j < Lm} (w^\wedge(k)_j)^2 / m - L \right). \quad (2.2)$$

**Proposition 2.1** (Hou-Zhao [7], Lemma 2.1). *If a certificate satisfies (2.1) and  $b > 0$ , then*

$$h(N) \leq N^{1/2} + \sqrt{ab} \cdot N^{1/4} + O(1).$$

The uniform kernel with all boundary weights equal to 1 gives  $a = b = 1$  and recovers the Erdős-Turán-Lindström constant. Optimizing over certificates is what produced the constant 0.9435 of [7].

**Definition 2.2.**  $\gamma^*_{\text{sym}} := \inf \sqrt{ab}$ , the infimum being over all certificates at all resolutions  $(m, L, K)$ .

The infimum is non-increasing in  $L$  and in  $K$ , since surplus boundary weights may be set to 1 and surplus kernels given weight 0, and it is non-increasing under the refinement  $m \rightarrow 2m$ , since splitting each bin in two preserves (2.1) exactly and leaves (2.2) unchanged. It is not a priori clear that  $\gamma^*_{\text{sym}} > 0$ ; Theorem A gives the first proof that it is, together with an explicit bound. As a by-product the hypothesis  $b > 0$  in Proposition 2.1 is automatic: Theorem 3.4 gives  $b \geq 1/(2a) > 0$  for every certificate satisfying (2.1).

### 2.2 The continuum relaxation

Let  $K \geq 1$  and let  $\rho_1, \dots, \rho_K$  be symmetric probability densities on  $[0, 1]$ ; let  $\omega_1, \dots, \omega_K : [0, \infty) \rightarrow \mathbb{R}$  be measurable with  $\omega_k \equiv 1$  outside a compact set and  $\omega_k - 1 \in L^2$ ; and let  $\lambda_k \geq 0$  with  $\sum \lambda_k = 1$ . Call this data a *continuum certificate* if

$$\sum_k \lambda_k \int_0^1 \rho_k(x) \omega_k(u+x) dx \geq 1 \quad \text{for all } u \geq 0, \quad (2.3)$$

and set

$$a = \sum_k \lambda_k \int_0^1 \rho_k(x)^2 dx, \quad b = 1 + 2 \sum_k \lambda_k \int_0^\infty (\omega_k(y)^2 - 1) dy. \quad (2.4)$$

**Lemma 2.3.** *Every discrete certificate at resolution  $(m, L, K)$  is a continuum certificate with the same values of  $a$  and  $b$ .*

*Proof.* Read  $p^\wedge(k)$  and  $w^\wedge(k)$  as step functions on cells of width  $1/m$ : put  $\rho_k = m \cdot p^\wedge(k)_i$  on  $[i/m, (i+1)/m)$  and  $\omega_k = w^\wedge(k)_j$  on  $[j/m, (j+1)/m)$ , with  $\omega_k \equiv 1$  beyond  $Lm/m$ . Then  $\rho_k$  is a symmetric probability density on  $[0, 1]$ ,  $\omega_k - 1$  has compact support, and the expressions (2.4) reduce to (2.2). Let  $C_q$  denote the left-hand side of the  $q$ -th inequality in (2.1), with the convention  $C_q = 1$  for  $q > Lm$ , which is consistent because  $\sum_i p^\wedge(k)_i = 1$ . For a shift  $u = (q+s)/m$  with  $q \in \mathbb{Z}_{\geq 0}$  and  $0 \leq s < 1$ , the left-hand side of (2.3) is piecewise linear in  $s$  and equals

$$(1-s) \cdot C_q + s \cdot C_{q+1} \geq 1,$$

since each of  $C_q$  and  $C_{q+1}$  is at least 1. Hence (2.3) holds for every  $u \geq 0$ . ■

Consequently any lower bound proved for the continuum problem bounds every discrete certificate at every finite resolution, with no limiting argument and no separate continuum solver. Write  $\gamma^*_{\text{cont}}$  for the infimum of  $\sqrt{ab}$  over continuum certificates; Lemma 2.3 gives  $\gamma^*_{\text{cont}} \leq \gamma^*_{\text{sym}}$ , and Theorem A is proved for  $\gamma^*_{\text{cont}}$ .

## 2.3 Two-sided certificates

A *two-sided certificate* at resolution  $(m, L, K)$  consists of kernels  $p^\wedge(k) \in \mathbb{R}^m$  that are nonnegative with  $\sum_i p^\wedge(k)_i = 1$  but need not be symmetric, mixing weights  $\lambda$  as before, and for each  $k$  two boundary vectors  $u^\wedge(k), v^\wedge(k) \in \mathbb{R}^{(Lm)}$ , extended by 1 for  $j \geq Lm$ , subject to

$$\sum_k \lambda_k \sum_i p^\wedge(k)_i u^\wedge(k)_{q+i} \geq 1 \quad (q = 0, \dots, Lm),$$

$$\sum_k \lambda_k \sum_i p^\wedge(k)_{m-1-i} v^\wedge(k)_{q+i} \geq 1 \quad (q = 0, \dots, Lm), \quad (2.5)$$

with  $a$  as in (2.2) and

$$b = 1 + (1/m) \sum_k \lambda_k \sum_{j < Lm} [(u^\wedge(k)_j)^2 + (v^\wedge(k)_j)^2 - 2]. \quad (2.6)$$

A symmetric certificate with  $u = v = w$  is a two-sided certificate with the same  $a$  and  $b$ .

**Proposition 2.4.** *If a two-sided certificate satisfies (2.5) and  $b > 0$ , then  $h(N) \leq N^{1/2} + \sqrt{ab} \cdot N^{1/4} + O(1)$ .*

*Proof.* We follow the proof of Theorem 2.1 in [7] and indicate the one change. There the weight function  $Q_r$  equals the boundary vector  $w^\wedge(r)$  on the left boundary region and its reflection on the right one, and the covering claim (6) of [7] is checked separately near each end. Symmetry of the kernel is used only in the right-hand case: under the substitution  $s' = H - 1 - s$ , block  $i$  of the smoothing kernel  $K^\wedge(r)$  becomes block  $m - 1 - i$ , and symmetry is what lets the right-hand calculation repeat the left-hand one. Define  $Q_r$  instead by  $u^\wedge(r)$  on the left region and by the reflection of  $v^\wedge(r)$  on the right. The left-hand calculation is unchanged and yields the first system in (2.5); the right-

hand calculation is the same computation for the reversed vector  $(\hat{p}(r)_{m-1-i})_i$  against  $\hat{v}(r)$ , and yields the second. The energy bound (5) of [7] uses only that the autocorrelation of  $\hat{K}(r)$  is even, which holds for every kernel, and the computation of  $\|Q_r\|_{2^2}$  now gives  $N + H - 1 + h \sum_j [(\hat{u}(r)_j)^2 - 1] + h \sum_j [(\hat{v}(r)_j)^2 - 1]$ , whose  $\lambda$ -average is  $N + bH - 1$  with  $b$  as in (2.6). The rest of the proof is verbatim. ■

**Definition 2.5.**  $\gamma^* := \inf \sqrt{ab}$  over all two-sided certificates at all resolutions. Thus  $\gamma^* \leq \gamma^*_{\text{sym}}$ .

Lemma 2.3 applies to each end separately. A two-sided certificate is therefore a two-sided continuum certificate — kernels  $\rho_k$  on  $[0,1]$  that need not be symmetric, profiles  $\omega_k$  and  $\omega'_k$  for the two ends, with (2.3) required for  $\rho_k$  against  $\omega_k$  and for the reflection  $\bar{\rho}_k(x) = \rho_k(1 - x)$  against  $\omega'_k$ , and  $b = 1 + \sum_k \lambda_k \int_0^\infty (\omega_k^2 + \omega'_k{}^2 - 2)$  — with the same  $a$  and  $b$ .

### 3. A lower bound for the certifiable constant

Throughout this section  $(\lambda, \rho, \omega)$  is a continuum certificate. For each  $k$  write  $R_k$  for the distribution function of  $\rho_k$ ,

$$F_k = \int_0^1 \rho_k(x)^2 dx = \|\rho_k\|_{2^2}, \quad G_k = E|X_k - Y_k|, \quad (3.1)$$

where  $X_k, Y_k$  are independent with density  $\rho_k$ , so that  $G_k$  is the Gini mean difference of  $\rho_k$ , and  $a = \sum_k \lambda_k F_k$ . Put  $e_k = \omega_k - 1$ , so that

$$b = 1 + \sum_k \lambda_k \int_0^\infty (4e_k + 2e_k^2) dt, \quad \sum_k \lambda_k \int_0^1 \rho_k(x) e_k(u+x) dx \geq 0 \quad (u \geq 0). \quad (3.2)$$

By Cauchy-Schwarz on  $[0, 1]$ ,  $F_k \geq (\int_0^1 \rho_k)^2 = 1$ , so  $a \geq 1$  with equality precisely for uniform kernels. This alone gives only  $\gamma^2 \geq b$ , which is vacuous without a lower bound on  $b$ ; the content of the argument is the trade-off between  $a$  and  $b$ .

#### 3.1 The flat multiplier

**Lemma 3.1.** *Every continuum certificate satisfies  $b \geq \sum_k \lambda_k G_k$ .*

*Proof.* For fixed kernels, minimizing  $b$  subject to (3.2) is a convex quadratic program in  $(e_k)$ , so integrating the constraint against any nonnegative measure and subtracting from the objective gives a valid lower bound. Take the multiplier  $4 \cdot (\text{Lebesgue measure on } [0, \infty))$ . By Fubini,  $\int_0^\infty \int_0^1 \rho_k(x) e_k(u+x) dx du = \int_0^\infty e_k(t) R_k(t) dt$ , whence

$$b \geq 1 + \sum_k \lambda_k \int_0^\infty [2e_k(t)^2 + 4e_k(t)(1 - R_k(t))] dt.$$

The integrand is minimized pointwise at  $e_k(t) = -(1 - R_k(t))$ , with value  $-2(1 - R_k(t))^2$ . As  $R_k \equiv 1$  on  $[1, \infty)$ , the integral is over  $[0, 1]$  only, so  $b \geq \sum_k \lambda_k [1 - 2\int_0^1 (1 - R_k)^2]$ . Since  $\rho_k$  is symmetric on  $[0, 1]$ ,  $\int_0^1 (1 - R_k) = E[X_k] = 1/2$ , hence

$$1 - 2\int_0^1 (1 - R_k)^2 = 2\int_0^1 (1 - R_k) - 2\int_0^1 (1 - R_k)^2 = 2\int_0^1 R_k(1 - R_k) = 2 E[(Y_k - X_k)_+] = G_k. \quad \blacksquare$$

#### 3.2 Mixtures collapse

**Lemma 3.2.**  $a \cdot b \geq \min_k F_k G_k$ .

*Proof.* By Lemma 3.1 and Cauchy-Schwarz applied to the weights  $\lambda$ ,

$$a \cdot b \geq (\sum_k \lambda_k F_k)(\sum_k \lambda_k G_k) \geq (\sum_k \lambda_k \sqrt{F_k G_k})^2 \geq \min_k F_k G_k. \quad \blacksquare$$

Mixing kernels is therefore powerless against the lower bound, although it is not

powerless for the upper bound: numerically a second kernel buys  $2.9 \times 10^{-3}$  in  $\gamma$  (Section 4.3). It is enough to bound the single-kernel quantity  $F \cdot G = Q(\rho)$  from below, where

$$Q(\rho) := \|\rho\|_{2^2}^2 \cdot E|X - Y|, \quad X, Y \text{ independent with density } \rho. \quad (3.3)$$

$Q$  is invariant under translation and under the scaling  $\rho \mapsto \rho(\cdot/s)/s$ , so a sharp lower bound for it is a statement about the shape of  $\rho$  alone.

### 3.3 A sharp scale-invariant inequality

**Theorem 3.3.** *For every probability density  $\rho$  on  $\mathbb{R}$  with  $\rho \in L^2$ ,*

$$Q(\rho) \geq \pi^2/32 = 0.3084251\dots, \quad (3.4)$$

*with equality precisely for the cosine arches  $\rho_W(x) = (\pi/4W) \cdot \cos(\pi x/2W)$  on  $[-W, W]$ ,  $W > 0$ , and their translates.*

*Proof.* Let  $F$  be the distribution function of  $\rho$  and let  $x(u) = \inf\{x : F(x) \geq u\}$ ,  $0 < u < 1$ , be its quantile function. Since  $\rho \in L^1 \cap L^2$  has no atoms,  $F$  is continuous and  $x$  is strictly increasing with  $F(x(u)) = u$ ; its derivative  $g = x'$  exists almost everywhere and satisfies  $\rho(x(u)) \cdot g(u) = 1$  for almost every  $u$ . The substitution  $u = F(x)$  gives

$$\|\rho\|_{2^2}^2 = \int \rho(x)^2 dx = \int_0^1 \rho(x(u)) du = \int_0^1 du / g(u). \quad (3.5)$$

Next,  $E|X - Y| = 2 \int F(x)(1 - F(x)) dx$ . Writing this integral as a Lebesgue-Stieltjes integral in  $u$  and discarding the singular part of  $dx$  — which is supported on the gaps of the support of  $\rho$ , across which  $x$  jumps, and which contributes nonnegatively — gives

$$E|X - Y| = 2 \int_0^1 u(1 - u) dx(u) \geq 2 \int_0^1 u(1 - u) g(u) du. \quad (3.6)$$

By Cauchy-Schwarz, and since  $\int_0^1 \sqrt{u(1-u)} du$  is the area of a half-disc of radius  $1/2$ ,

$$\left( \int_0^1 du/g(u) \right) \left( \int_0^1 u(1-u) g(u) du \right) \geq \left( \int_0^1 \sqrt{u(1-u)} du \right)^2 = (\pi/8)^2. \quad (3.7)$$

Multiplying (3.5) and (3.6) and applying (3.7) gives  $Q(\rho) \geq 2(\pi/8)^2 = \pi^2/32$ . Equality forces the singular part of  $dx$  to vanish and  $g(u) = C/\sqrt{u(1-u)}$  for some  $C > 0$ , so that  $x(u) = c + C \cdot \arcsin(2u - 1)$  and  $F(x) = 1/2(1 + \sin((x - c)/C))$  for  $|x - c| \leq \pi C/2$ ; this is the cosine arch  $\rho_W$  with  $W = \pi C/2$ , translated to  $c$ . Conversely  $\|\rho_W\|_{2^2}^2 = \pi^2/(16W)$  and  $E|X - Y| = W/2$ , so  $Q(\rho_W) = \pi^2/32$ . ■

**Remark 3.3.1.** The cosine arch is also the stationary point of  $Q$ : on its support it satisfies the Euler-Lagrange equation  $\rho(x) = c + \lambda \int |x - y| \rho(y) dy$  with  $\lambda = -\pi^2/(8W^2)$ , since differentiating twice turns that equation into  $\rho'' = 2\lambda\rho$ . Stationarity by itself does not prove (3.4). Normalizing  $E|X - Y| = 1$  by scaling, the problem becomes the minimization of the convex functional  $\|\rho\|_{2^2}^2$  over  $\{E|X - Y| \leq 1\}$ , and since  $\rho \mapsto E|X - Y|$  is concave on densities, that constraint set is not convex; the sign of the multiplier above is the visible symptom. The quantile argument avoids the issue and delivers the equality case directly.

**Remark 3.3.2.** Inequality (3.4) is a statement about the Gini mean difference and the  $L^2$  norm of a density, and is elementary; we have not located it in the literature in this form, and make no claim of novelty for it.

*Figure 1. The functional  $Q$  of (3.3) on standard densities, with the bound  $\pi^2/32$  of Theorem 3.3 marked. The cosine arch attains it; every other density lies strictly above.*

### 3.4 One atom at the origin

**Theorem 3.4.** *Every continuum certificate satisfies*

$$b \geq \sum_k \lambda_k G_k + 1/(2a) \quad \text{and} \quad a \cdot b \geq (\pi^2 + 16)/32 = 0.8084251\dots \quad (3.8)$$

*Proof.* Apply weak duality as in Lemma 3.1, now with the multiplier  $\mu = 4 \cdot \text{Leb} + \beta \cdot \delta_0$  for a parameter  $\beta \geq 0$ ; the atom is admissible because the constraint in (3.2) at  $u = 0$  is one of the constraints. This gives

$$b \geq 1 + \sum_k \lambda_k \int_0^\infty [2e_k(t)^2 + 4e_k(t) - e_k(t)(4R_k(t) + \beta \rho_k(t))] dt,$$

and pointwise minimization in  $e_k(t)$  completes the square with value  $-(1/8)[4(1 - R_k(t)) - \beta \rho_k(t)]^2$ , so that

$$b \geq 1 - (1/8) \sum_k \lambda_k \int_0^\infty [16(1 - R_k)^2 - 8\beta(1 - R_k)\rho_k + \beta^2 \rho_k^2] dt.$$

The cross term is the kernel-independent constant (1.3), and the two remaining terms are  $G_k$  and  $F_k$  as in Lemma 3.1 and (3.1). Hence

$$b \geq \sum_k \lambda_k [G_k + \beta/2 - (\beta^2/8) F_k] = \sum_k \lambda_k G_k + \beta/2 - (\beta^2/8) \cdot a,$$

for every  $\beta \geq 0$ . The right-hand side is a concave quadratic in  $\beta$ , maximized at  $\beta^* = 2/a$ , which gives the first claim. Multiplying by  $a$  and using Lemma 3.2 and Theorem 3.3,

$$a \cdot b \geq (\sum_k \lambda_k F_k)(\sum_k \lambda_k G_k) + 1/2 \geq \min_k Q(\rho_k) + 1/2 \geq \pi^2/32 + 1/2. \blacksquare$$

For symmetric certificates Theorem A follows by Lemma 2.3, since  $\gamma = \sqrt{ab} \geq \sqrt{((\pi^2 + 16)/32)} = 0.89912465\dots$  for every continuum certificate and hence for every discrete one. Taking only the flat part of the multiplier gives the weaker bound  $\gamma \geq \pi/(4\sqrt{2}) = 0.555360\dots$

**Theorem 3.5.** *Every two-sided continuum certificate satisfies  $b \geq \sum_k \lambda_k G_k + 1/(2a)$ , and hence  $a \cdot b \geq (\pi^2 + 16)/32$ .*

*Proof.* Apply the argument of Theorem 3.4 at each end separately, with the multiplier  $2 \cdot \text{Leb} + \beta' \cdot \delta_0$  at each. The first end involves  $\rho_k$  and the second its reflection  $\bar{\rho}_k$ , whose distribution function is  $1 - R_k(1 - x)$ . Completing the square at each end and adding, the two quadratic terms combine to  $\int_0^1 [(1 - R_k)^2 + R_k^2] = 1 - G_k$  with no symmetry assumption, and each cross term equals  $1/2$  by (1.3). Hence  $b \geq \sum_k \lambda_k G_k + \beta' - (\beta'^2/2) \cdot a$  for every  $\beta' \geq 0$ , and  $\beta' = 1/a$  gives the first claim. Lemma 3.2 and Theorem 3.3 complete the proof as before.  $\blacksquare$

Theorem A therefore holds for  $\gamma^*$  and not only for  $\gamma^*_{\text{sym}}$ . In the symmetric case symmetry enters Lemma 3.1 only through the identity  $\int_0^1 (1 - R_k) = 1/2$ ; the two-sided argument does not need it, because between them the two ends see  $R_k$  and its reflection. As a by-product the hypothesis  $b > 0$  in Proposition 2.4 is automatic as well.

### 3.5 Sharpness, and where the remaining gap sits

Lemma 3.2 and Theorem 3.3 are sharp: the minimizing cosine arch is symmetric and supported on an interval, so after rescaling to  $[0, 1]$  it is an admissible kernel, and a single kernel makes Lemma 3.2 an equality. The final inequality of Theorem 3.4 is sharp in the same sense, so (3.8) is the best bound the multiplier family  $\{4 \cdot \text{Leb} + \beta \cdot \delta_0\}$  can give. The gap between 0.899125 and the numerical floor 0.9428091 therefore lies entirely in the choice of multiplier. For fixed kernels the inner problem is a convex quadratic program whose exact value is the supremum over all nonnegative multipliers, and the flat and one-atom families are the two simplest members of that supremum. Table 1 quantifies the loss on two reference kernels.

| kernel              | F·G (flat dual)     | F·G + $\frac{1}{2}$ (one-atom dual) | F·B (exact inner value) |
|---------------------|---------------------|-------------------------------------|-------------------------|
| uniform on $[0, 1]$ | $1/3 = 0.3333$      | $5/6 = 0.8333$                      | 1.0000                  |
| cosine arch         | $\pi^2/32 = 0.3084$ | $(\pi^2+16)/32 = 0.8084$            | 0.9820                  |

Table 1. Lower bounds on  $a \cdot b$  for a single kernel, as  $F$  times the corresponding bound on  $b$ , against the exact value  $F \cdot B$  of the inner quadratic program. For the uniform kernel  $B = 1$  exactly: summing the constraint of (3.2) over integer shifts gives  $\sum_{e \in \mathbb{Z}} e \geq 0$ , whence  $b \geq 1$ , with equality at  $e \equiv 0$ . The value 0.9820 for the cosine arch is numerical, computed on grids of 64, 128 and 256 bins. For the two kernels of the  $m = 512$  certificate of Section 4 the one-atom bound gives 0.869 and 0.833 against exact inner values 0.922 and 0.959.

## 4. The floor, numerically

This section reports the numerical side: resolution scans in both settings with boundary depth  $L = 4$ , the effect of asymmetry, and the structure of the limiting certificates. Section 5.2 revisits the upper bound with an explicit certificate and a deeper boundary layer.

### 4.1 Method

The problem is bilinear: for fixed kernels and mixing weights the optimization over boundary weights is a convex quadratic program, and for fixed boundary weights the optimization over kernels is again convex, but the joint problem is not. Alternating minimization stalls at  $\gamma \approx 0.982$ , a poor stationary point. The scheme used here is a nested one:

- **Inner.** For fixed  $(p, \lambda)$ , minimize  $b$  over the boundary weights subject to the covering inequalities. This is a quadratic program with  $Lm$  variables per kernel and  $Lm + 1$  constraints; for two-sided certificates it splits into two independent programs, one for each end. We solve each through its nonnegative dual [7, Section 3.3], whose matrix is banded with bandwidth  $m$ , by a primal-dual active-set iteration warm-started from the previous outer step.
- **Outer.** L-BFGS over the kernels — softmax-parametrized to enforce nonnegativity and normalization, and in the symmetric case reduced to  $m/2$  free parameters per kernel — and over the mixing weights, with the gradient of the inner value obtained from the quadratic-program duals by the envelope theorem.
- **Globalization.** Multi-start from 30 or more random seeds at  $m = 32$ , with basin hopping; each higher resolution is initialized by upsampling the converged certificate of the previous one, splitting every bin in two. Splitting preserves feasibility and leaves  $\gamma$  unchanged, but it also produces an exact stationary point of the finer problem — the gradient vanishes there — so the upsampled certificate must be perturbed before it is re-optimized, or the optimizer stops where it starts.

The analytic gradient agrees with central finite differences to  $5.5 \times 10^{-6}$ . As a check on the implementation, at the resolution used in [7] —  $m = 32$ ,  $L = 4$ , a single kernel — the optimizer returns  $\gamma = 0.946147$ , the value reported there for that case.

An independent implementation of the two-sided optimizer, with the inner problem solved as above, reproduces the single-kernel value 0.9461473 of [7] and the symmetric values of Table 2, and its gradient agrees with central finite differences to a relative  $3 \times 10^{-8}$ .

Every value reported below is verified independently of the optimizer and of floating-

point arithmetic. The converged certificate is rationalized, the boundary program is resolved for the rational kernels and mixing weights, the boundary weights are rounded, a uniform increment restores any covering deficit [7, Lemma 3.2], all inequalities of (2.1) or (2.5) are checked over  $\mathbb{Q}$ , and  $a$ ,  $b$  and  $\gamma^2$  are evaluated exactly; at  $m = 256$  a two-sided certificate involves 2050 inequalities and at  $m = 512$  a symmetric one 2049. A verified value is an upper bound for  $\gamma^*$  by Proposition 2.1 and Definition 2.2, whatever the quality of the optimization that produced it.

## 4.2 Resolution scan and extrapolation

| <b>m</b> | <b>symmetric <math>\gamma</math><br/>(verified)</b> | <b>two-sided <math>\gamma</math><br/>(verified)</b> | <b>two-sided drop</b> | <b>ratio</b> |
|----------|-----------------------------------------------------|-----------------------------------------------------|-----------------------|--------------|
| 32       | 0.94324229                                          | 0.9429206688                                        | —                     | —            |
| 64       | 0.94292072                                          | 0.9428374024                                        | $8.33 \times 10^{-5}$ | —            |
| 128      | 0.94283748                                          | 0.9428162103                                        | $2.12 \times 10^{-5}$ | 3.93         |
| 256      | 0.94281628                                          | 0.9428108643                                        | $5.35 \times 10^{-6}$ | 3.96         |
| 512      | 0.9428108915                                        | —                                                   | —                     | —            |

Table 2. Verified values at  $L = 4$ ,  $K = 2$ . The symmetric entries up to  $m = 256$  are those of the original scan; the  $m = 512$  entry is the re-optimized certificate of Section 4.1 (the original  $m = 512$  certificate gave 0.94281112). A two-sided certificate at resolution  $m$  does as well as a symmetric one at  $2m$ , and slightly better. The two-sided drops fall by factors of 3.93 and 3.96, consistent with convergence in  $1/m^2$ .

For the two-sided scan, Richardson extrapolation in  $1/m^2$  gives 0.9428091 from  $m = 64$  and 128 and 0.94280908 from  $m = 128$  and 256, and three-point fits on  $m = 64$ , 128 and 256 that add a  $1/m^3$  or a  $1/m^4$  term give 0.94280907 and 0.94280908. For the symmetric scan, Richardson from  $m = 256$  and 512 gives 0.94280910. (The original, under-converged  $m = 512$  value gave 0.9428094 instead; summing the tail geometrically with ratio  $1/4$  is algebraically identical to the last Richardson step and is not an independent estimate.) For the infimum  $\gamma^*_4$  over certificates with  $L = 4$  we record

$$\gamma^*_4 = 0.9428091 \pm 1 \times 10^{-7}, \quad (4.1)$$

the uncertainty covering the spread of the extrapolations in both scans rather than a rigorous error bound; the two-sided Richardson estimate for  $\gamma^*_4$  is 0.88888897. Section 5.2 shows that a deeper boundary layer reaches  $2\sqrt{2}/3 = 0.94280904\dots$ , about  $5 \times 10^{-8}$  lower.

Figure 2. The verified values of Table 2 against  $1/m^2$ , symmetric and two-sided, with the extrapolated floor marked.

## 4.3 Saturation in the other two parameters

The boundary depth and the number of kernels both saturate, so that  $m$  is the only parameter whose refinement matters. At  $m = 32$  and  $K = 2$ , taking  $L = 3$  gives 0.943245 while  $L = 4, 5$  and 6 all give 0.943242: the boundary layer is under three kernel widths deep. With  $L = 4$ , a single kernel gives 0.946147, two kernels give 0.943242, and three kernels give 0.943242 again — identical to five decimals at both  $m = 32$  and  $m = 64$ , with the optimizer driving one of the three weights down to about 0.04. The framework’s capacity saturates at two kernels, which buy  $2.9 \times 10^{-3}$  over one. The saturation in  $L$  is a property of these symmetric optima; it fails for the extremal two-sided certificate of Section 5.2, whose boundary profile needs about six kernel widths, and which loses about  $10^{-4}$  in  $b$  when confined to  $L = 4$ .

## 4.4 The limiting certificate

The  $m = 512$  certificate of the original symmetric scan is a two-kernel mixture with  $\lambda = (0.3396, 0.6604)$ ; re-optimization changes this only slightly, to  $(0.3383, 0.6617)$ . The kernel carrying weight 0.34 is narrow and spiked, with peak density 2.69 and a peak-to-edge ratio of 17.3; the kernel carrying weight 0.66 is broad, with peak 1.87 and ratio 9.0. Both are symmetric and unimodal.

The boundary profiles are the more informative half. They dip to roughly 0.08–0.10 at the outermost cell, overshoot above 1 near half a kernel width — reaching 1.37 — and oscillate back to 1 by about three kernel widths. The overshoot is what allows the covering inequality (2.1) to hold while the outermost cells are heavily discounted. That the profile is confined to a few kernel widths is the dual-side statement that an argument of this shape sees only a boundary layer of width  $O(N^{3/4})$  — the same scale that appears in Cilleruelo’s theorem on gaps in dense Sidon sets [4], reached here independently.

The two-sided optimum is organized differently. Its mixing weights are  $\lambda \approx (0.15, 0.85)$  at every resolution from  $m = 32$  to 256; the lighter kernel is strongly asymmetric, with total variation distance 0.67 between it and its reversal, while the heavier one is close to symmetric, at 0.19.

*Figure 3. The limiting certificate at  $m = 512$ : the two kernel densities (left) and their boundary weight profiles (right, in units of kernel widths).*

## 4.5 The value $2\sqrt{2}/3$

The extrapolated value (4.1) is close to

$$2\sqrt{2}/3 = 0.94280904\dots, \quad \text{equivalently} \quad \gamma^{*2} = 8/9, \quad (4.2)$$

which lies  $3\text{--}4 \times 10^{-8}$  below every two-sided extrapolation computed above; extrapolating  $\gamma^2$  instead of  $\gamma$  gives 0.88888897 against  $8/9 = 0.88888889$ . The symmetric scan, once its  $m = 512$  value is re-converged, points to the same place (0.94280910). The agreement is closer than the spread between estimates from different pairs of resolutions, which makes it suggestive, but it is not decisive.

At this precision the question cannot be settled by refining the  $L = 4$  scans: under a  $1/m^2$  model a two-sided certificate at  $m = 512$  returns about 0.9428095 whether  $\gamma^*_4$  equals  $2\sqrt{2}/3$  or 0.9428091. Section 5.2 settles the upper half by a different route. An explicit two-sided certificate — the linear kernel  $\rho(x) = 2x$  — attains  $a \cdot b = 8/9$  in the continuum, and approaches it at resolution  $m$  with  $a \cdot b - 8/9 \approx 2/(9m^2)$  once  $L \geq 6$ ; the residual  $3\text{--}4 \times 10^{-8}$  in the  $L = 4$  scans is the cost of the four-width box. Whether anything goes below  $2\sqrt{2}/3$  is the lower-bound question of Section 5.4.

## 4.6 Asymmetry and the floor

Proposition 2.4 admits asymmetric kernels, and it is natural to ask whether they lower the floor. Table 3 compares symmetric and two-sided certificates with one and two kernels.

| <b>L = 4</b>     | <b>m = 32</b> | <b>64</b> | <b>128</b> | <b>256</b> | <b>limit</b>       |
|------------------|---------------|-----------|------------|------------|--------------------|
| symmetric, K = 1 | 0.946147      | 0.945669  | 0.945610   | —          | $\approx 0.94559$  |
| two-sided, K = 1 | 0.942944      | 0.942855  | 0.942832   | —          | $\approx 0.942825$ |
| symmetric, K = 2 | 0.943242      | 0.942921  | 0.942837   | 0.942816   | 0.9428091          |
| two-sided, K = 2 | 0.942921      | 0.942837  | 0.942816   | 0.942811   | 0.9428091          |

Table 3. Symmetric against two-sided certificates. The  $K = 2$  rows and the  $m = 32$  entries are verified values; the remaining  $K = 1$  entries are optimizer values. Limits are Richardson extrapolations from the last two resolutions shown; for the symmetric single kernel the convergence is irregular and the limit is rough.

For a single kernel, asymmetry lowers the limit by almost  $3 \times 10^{-3}$ , to within  $2 \times 10^{-5}$  of the two-kernel floor: the single-kernel value 0.946147 of [7] is a property of the symmetric parametrization rather than of one kernel as such. For two kernels, asymmetry does not move the limit. A two-sided certificate at resolution  $m$  does as well as a symmetric one at  $2m$  — slightly better, by  $5\text{--}8 \times 10^{-8}$ , at  $m = 32, 64$  and  $128$  — and the two scans extrapolate to the same value. Mixing and asymmetry are two different ways of reaching the same floor, and combining them gains nothing in the limit. We take this as evidence that 0.94281 is a property of the smoothing argument itself rather than of the way its certificates are parametrized. The benefit of asymmetry is computational: it halves the resolution needed for a given accuracy. All of this is at  $L = 4$ . With a deeper boundary layer a single asymmetric kernel reaches the floor on its own (Section 5.2), so the two-kernel mixtures found at  $L = 4$  are best read as approximations to it within the four-width box.

A second natural variation, decoupling the grid of the kernel from that of the boundary weights, cannot move  $\gamma^*$  at all. A certificate whose kernel and boundary weights live on grids of different spacing is a special case of an ordinary certificate on the common refinement of the two grids, with the same  $a$  and  $b$ ; a kernel narrower than a block is already allowed, since  $p$  may vanish on part of  $[0,1]$ ; and the continuum relaxation of Section 2.2 has no grid.

#### 4.7 Caveats

- The value of  $\gamma^*$  is not proved. Theorem B and Theorem 5.3 are rigorous, but the conjecture  $\gamma^* = 2\sqrt{2}/3$  rests on the optimizer having found the global optimum at each resolution, at  $L = 4$  and at  $L = 8$ , and on the convergence of the discrete values. Every restart from random seeds at  $m = 32$ , and every perturbed upsampled start at higher  $m$ , landed in the same basin.
- The optimum has flat directions. At  $m = 256$  the constants  $a$  and  $b$  of the two-sided certificate change by about  $10^{-3}$  over the last few hundred iterations while  $\gamma$  changes by  $10^{-9}$ , and at  $m = 32$  symmetric optima with quite different splits  $\lambda$  attain the same  $\gamma$ . Only  $\gamma$  is well determined; the individual values of  $a$  and  $b$  in Appendix A should not be read as converged.
- Only the implication to  $h(N)$  depends on Propositions 2.1 and 2.4; Theorems A and B do not.

## 5. The optimal multiplier, an extremal certificate, and open questions

## 5.1 The optimal dual multiplier

For fixed kernels the boundary problem is a quadratic program, and its dual solution is a nonnegative measure  $\mu$  on the shifts  $u \geq 0$  — the multiplier of Section 3, now optimal rather than chosen. At the optimal certificates of Section 4 every covering constraint is active, and the optimality conditions then read

$$(\mu * R)(u) = \int_0^\infty R(u - v) d\mu(v) = 4 \quad \text{for all } u \geq 0, \quad (5.1)$$

where  $R = \sum_k \lambda_k p_k * \tilde{p}_k$  is the mixed autocorrelation of the kernels,  $\tilde{p}_k(x) = p_k(-x)$ , and the normalization is that of Section 3, in which the flat multiplier is 4·Leb. For two-sided certificates each end has its own multiplier, and both satisfy (5.1) with the same  $R$  (Lemma 5.4). Equation (5.1) is a Wiener-Hopf equation of the first kind. The one-atom multiplier of Theorem 3.4 is its solution at  $u = 0$  when the continuous part is flat:  $4 \cdot \frac{1}{2} + \beta \cdot R(0) = 4$  gives  $\beta = 2/a$ .

Figure 4 (left) shows the optimal multiplier at the two-sided certificates of Section 4. As  $m$  grows it converges, at rate  $1/m$ , to

$$\mu^* = 2 \sum_{n \geq 0} U^{*n}, \quad (5.2)$$

twice the renewal measure [6, Chapter XI] of the uniform distribution  $U$  on  $[0,1]$ . This is an atom of mass 2 at the origin, density  $2e^u$  on  $[0,1)$ , density  $2(e^u - u e^{(u-1)})$  on  $[1,2)$ , and so on, converging to the flat value 4 with oscillations that decay like  $e^{-2.09u}$ . At resolution  $m$  the atom has mass  $2 - 2/m$ , and the density just after it is  $2(1 + 1/m)$ , to the printed digits for  $m = 32, 64$  and  $128$ ; the mean deviation from (5.2) on  $[0,1)$  is 0.010, 0.005 and 0.003 at  $m = 64, 128$  and  $256$ . The renewal identity  $\mu^* = 2\delta_0 + U * \mu^*$  explains the flat value: the bulk density is the atom, 2, divided by the mean  $\frac{1}{2}$  of  $U$ .

Truncations of (5.2) are the natural finite multiplier families. Table 4 shows how much each recovers at the optimal two-sided certificate at  $m = 256$ , measured against the gap between the flat-multiplier bound and the exact value; these are ceilings for what each family could prove, not proofs.

| multiplier                     | bound on $a \cdot b$ | share of the gap |
|--------------------------------|----------------------|------------------|
| flat, 4·Leb                    | 0.3748               | 0%               |
| flat + atom at 0 (Theorem 3.4) | 0.8580               | 94.00%           |
| + linear ramp on $[0,1)$       | 0.88809              | 99.84%           |
| + linear piece on $[1,2)$      | 0.88876              | 99.97%           |
| exact (optimal multiplier)     | 0.88889              | 100%             |

Table 4. Multiplier families evaluated at the optimal two-sided certificate ( $m = 256$ ,  $L = 4$ ), whose exact value is  $a \cdot b = 0.888892$ . Each family is fitted by maximizing the dual value over its parameters.

We also traced the optimum at prescribed  $a$ , minimizing  $b$  subject to  $R(0) = a$ . The product  $a \cdot b$  is remarkably flat along this curve: it changes by less than  $10^{-4}$  as  $a$  runs from 1.32 to 2.0, while the second kernel's weight falls below 0.01 above  $a \approx 1.5$ . The density just after the atom stays at half the flat value throughout, but the atom mass (about  $2.6/a$  at  $m = 64$ ) and the rest of the multiplier follow no simple rule in  $a$ ; the renewal form (5.2) appears only at the optimum. This flatness is also why  $a$  and  $b$  are poorly determined individually (Section 4.7).

Figure 4. Left: the optimal dual multiplier at the two-sided certificates of Section 4 ( $m = 64$  and  $256$ , density divided by the flat value 4; the atom of mass about 2 at  $u = 0$  is not shown), against twice the renewal density of the uniform distribution on  $[0,1]$ . Right:

the boundary weights of the certificate of Theorem 5.3 —  $e^t - 1$  on  $[0,1]$  at the end that sees  $\rho(x) = 2x$ , continued by the renewal convolution, and  $\omega \equiv 1$  at the other end.

## 5.2 An explicit certificate attaining $2\sqrt{2}/3$

The multiplier (5.2) determines the optimal autocorrelation through (5.1). By the renewal identity, (5.1) for  $u \geq 1$  follows from its restriction to  $[0,1]$ , and there it becomes a linear integral equation for  $R$  on  $[0,1]$  whose solution is

$$R^*(s) = (2/3)(1 - |s|)^2(2 + |s|), \quad |s| \leq 1. \quad (5.3)$$

This is the autocorrelation of the single linear density  $\rho(x) = 2x$  on  $[0,1]$ . That density is asymmetric, which is why symmetric certificates approach it only through mixtures.

**Theorem 5.3.** *Let  $\rho(x) = 2x$  on  $[0,1]$ , and let  $r$  be the renewal density of the uniform distribution on  $[0,1]$ . At the end whose covering condition involves the reflected kernel  $2(1 - x)$ , take  $\omega \equiv 1$ . At the other end take  $\omega = \frac{1}{4}(\mu * \rho)$ , that is,  $\omega(t) = e^t - 1$  for  $0 \leq t < 1$  and  $\omega(t) = \int_0^1 x \cdot r(t - x) dx$  for  $t \geq 1$ . This is a two-sided continuum certificate with  $a = 4/3$  and  $b = 2/3$ , so  $a \cdot b = 8/9$  and  $\gamma = 2\sqrt{2}/3$ .*

*Proof.* The reflected kernel is covered by  $\omega \equiv 1$  at no cost, and indeed  $\frac{1}{4}(\mu^* * \bar{\rho}) \equiv 1$ , so the constant profile is also optimal there. At the other end,  $\int_0^1 \rho(x) \omega(u + x) dx = \frac{1}{4}(\mu^* * R)(u)$ , and a direct computation with (5.3), using  $r(v) = e^v$  on  $[0,1]$  and  $r(v) = e^v - v$  on  $[1,2]$ , gives  $(\mu * R^*)(u) = 4$  for  $0 \leq u \leq 1$ ; the renewal identity extends this to every  $u \geq 0$ . Both covering conditions therefore hold with equality. Clearly  $a = \int_0^1 (2x)^2 dx = 4/3$ . For  $b$ , observe that for any multiplier  $\mu = 4 \cdot \text{Leb} + \nu$  on  $[0, \infty)$ , used at both ends, the argument of Theorem 3.5 gives

$$b \geq G + \langle \nu, T \rangle - \frac{1}{8} \langle \nu, R * \nu \rangle, \quad G = \int |s| R(s) ds, \quad T(\nu) = \int_{-\infty}^1 R(s) ds, \quad (5.4)$$

which reduces to Theorem 3.4 when  $\nu = \beta \delta_0$ . Here  $\mu^*$  is nonnegative and the profile above satisfies the optimality conditions with it, so (5.4) holds with equality for  $\mu = \mu^*$ . Under (5.1),  $\langle \nu, R * \nu \rangle = 4 \langle \nu, T \rangle$ , whence  $b = G + \frac{1}{2} \langle \nu, T \rangle$ . For (5.3),  $G = 4/15$  and  $\langle \nu, T \rangle = 2T(0) + \int_0^1 (2e^v - 4) T(v) dv = 4/5$ , so  $b = 4/15 + 2/5 = 2/3$ . Integrating  $\omega^2 - 1$  directly gives the same value. ■

The discrete versions behave exactly as the continuum predicts. At resolution  $m$  the kernel  $p_i = (2i + 1)/m^2$  has  $a = 4/3 - 1/(3m^2)$ , and with  $L = 8$  the exact boundary program gives  $b = 2/3 + 1/(3m^2)$  to within  $3 \times 10^{-8}$  for  $m \leq 512$  (Table 5). Hence  $a \cdot b = 8/9 + 2/(9m^2) - 1/(9m^4)$  to that accuracy, and the successive drops in  $\gamma$  fall by factors 3.998, 3.9994 and 3.9997. At  $L = 4$  the same kernel loses about  $10^{-4}$  in  $b$ , because its boundary profile needs about six kernel widths to settle (Figure 4, right); this is why the  $L = 4$  scans of Section 4 stop  $3\text{--}4 \times 10^{-8}$  above  $2\sqrt{2}/3$ . Optimizing freely at  $L = 8$  from this kernel lowers  $\gamma$  only slightly at finite  $m$  (by  $4 \times 10^{-7}$  at  $m = 64$ , and by less at  $m = 128$ ), with the kernel moving by less than 1% in  $L^1$ , consistent with a discretization gain that vanishes as  $m$  grows.

| <b>m</b> | <b><math>\gamma</math></b> | <b><math>b - 2/3</math></b> | <b><math>1/(3m^2)</math></b> |
|----------|----------------------------|-----------------------------|------------------------------|
| 32       | 0.9429240903               | $3.2555 \times 10^{-4}$     | $3.2552 \times 10^{-4}$      |
| 64       | 0.9428378328               | $8.1413 \times 10^{-5}$     | $8.1380 \times 10^{-5}$      |
| 128      | 0.9428162572               | $2.0377 \times 10^{-5}$     | $2.0345 \times 10^{-5}$      |
| 256      | 0.9428108625               | $5.1183 \times 10^{-6}$     | $5.0863 \times 10^{-6}$      |
| 512      | 0.9428095137               | $1.3036 \times 10^{-6}$     | $1.2716 \times 10^{-6}$      |

Table 5. The discretized linear kernel  $\rho(x) = 2x$  at  $L = 8$ , with the exact optimal boundary weights at each end. The certificate at  $m = 512$ , verified in exact rational

arithmetic, gives the bound  $\gamma < 0.942809522$  of Theorem B.

Theorem 5.3 proves  $\gamma^*_{\text{cont}} \leq 2\sqrt{2}/3$  for the continuum relaxation. For discrete certificates, Table 5 gives  $\gamma^* < 0.942809522$ , and a sequence of certificates converging to  $2\sqrt{2}/3$  at the rate  $2/(9m^2)$  in  $\gamma^2$ . We have not proved that the discrete values converge to  $2\sqrt{2}/3$ , so the statement  $\gamma^* \leq 2\sqrt{2}/3$  for the discrete framework rests on the numbers; they leave little room for doubt.

### 5.3 Two structural facts

**Lemma 5.4.** \*For two-sided certificates,  $b$  depends on the kernels and mixing weights only through the mixed autocorrelation  $R = \sum_k \lambda_k p_k * \tilde{p}_k$ , as does  $a = R(0)$ . For a single end the boundary cost also depends on the kernels' means, and these contributions cancel between the two ends.\*

*Proof sketch.* In the dual bound (5.4) for one end, the kernels enter in three ways. The first is through  $\langle \nu, R * \nu \rangle$ , which involves only  $R$ . The second is through  $\langle \nu, T \rangle$ , where  $T(\nu)$  is the probability that  $Y - X > \nu$  for independent  $X, Y$  with density  $\rho_k$ ;  $T$  is unchanged by reflection, since  $X$  and  $Y$  are exchangeable. The third is through the constant  $\int_0^1 (1 - R_k)^2$ , where  $R_k$  is the distribution function of  $\rho_k$ ; at the other end it becomes  $\int_0^1 R_k^2$ , and the sum of the two constants is  $1 - G_k$ , a function of  $R$ . Taking the supremum over  $\nu$  gives the claim. ■ Numerically, reversing any kernel of a two-sided certificate changes the sum of the two end costs by less than  $10^{-13}$ , while each end cost shifts by the change in the kernel's mean.

Consequently the two-sided problem is to minimize  $R(0) \cdot b(R)$  over the closed convex hull of autocorrelations of probability densities on  $[0,1]$ , with  $b$  a convex function of  $R$ , since it is a supremum of affine functionals. In this language Theorem 3.3 says  $R(0) \cdot \int |s| R(s) ds \geq \pi^2/32$ , and Theorem 5.3 says that the minimum is at most  $8/9$ , attained by (5.3).

### 5.4 Consequences and the lower bound

**Corollary 5.5.** *No certificate in the sense of Sections 2.1 or 2.3 proves a bound of the form  $h(N) \leq N^{1/2} + \gamma \cdot N^{1/4} + O(1)$  with  $\gamma < \sqrt{(\pi^2 + 16)/32} = 0.899124\dots$ ; and no such certificate proves an error term of size  $o(N^{1/4})$ . The best constant in this family is conjecturally  $2\sqrt{2}/3 = 0.94280904\dots$ , and  $0.942809522$  is attained by a verified certificate.*

The qualitative half of this statement — that arguments of this shape cannot approach (1.1) — follows already from the flat-multiplier bound  $\pi/(4\sqrt{2})$ ; what the atom adds is quantitative, and what the rest of the paper adds is the location of the true limit. The remaining room in this direction is small and mapped: about  $7 \times 10^{-4}$  below the constant of [7], and, if Conjecture 5.6 holds, less than  $5 \times 10^{-7}$  below the certificate of Theorem B. By Proposition 2.4 that certificate gives  $h(N) \leq N^{1/2} + 0.9428096 \cdot N^{1/4} + O(1)$ , the smallest constant certified so far, conditional as always on the smoothing lemma.

**What an improvement would have to retain.** The framework reaches its floor because it discards two quantities. The first is the variance term in Cilleruelo's identity for the number of representations, which is nonnegative and is thrown away when the energy is bounded; the second is the slack in the bound  $d(h) \leq 1$  on the multiplicity of small differences, which is far from tight for a set that is dense in an interval. Any argument that goes below  $\gamma^*$  must keep at least one of them, and both are quantities that the Cauchy-Schwarz step in this lineage is designed to avoid.

**Conjecture 5.6.**  $\gamma = 2\sqrt{2}/3$ , and the linear kernel  $p(x) = 2x$ , with the boundary weights of Theorem 5.3, is extremal.\*

Proving Conjecture 5.6 means proving  $a \cdot b \geq 8/9$  for every certificate. Two facts constrain how. First, no single multiplier can do it. For a fixed multiplier the bound on  $a \cdot b$  is the product of two positive affine functions of  $R$ , hence log-concave, so its minimum over all certificates is attained at a single kernel. Evaluating the optimal multiplier on other single kernels gives values far below  $8/9$ : on the lighter kernel of the  $L = 4$  two-kernel optimum it gives  $0.445$ . The multiplier must therefore depend on the certificate, as the atom  $2/a$  of Theorem 3.4 already does. Second, the target is now explicit: (5.2) and (5.3) are the optimal pair at the extremum. The truncations of Table 4, with parameters depending on  $a$  and perhaps on a few further moments of  $R$ , are the natural families. A proof along these lines needs a kernel-side inequality that generalizes Theorem 3.3 from the weight  $|s|$  to the weight that such a multiplier induces through (5.4).

## References

- [1] J. Balogh, Z. Füredi and S. Roy, An upper bound on the size of Sidon sets. *Amer. Math. Monthly* 130 (2023), no. 5, 437–445. arXiv:2103.15850.
- [2] T. F. Bloom, Erdős Problems, Problem #30. <https://www.erdosproblems.com/30>
- [3] D. Carter, Z. Hunter and K. O’Bryant, On the diameter of finite Sidon sets. *Acta Math. Hungar.* 175 (2025), no. 1, 108–126. arXiv:2310.20032.
- [4] J. Cilleruelo, Gaps in dense Sidon sets. *Integers* 0 (2000), #A11.
- [5] P. Erdős and P. Turán, On a problem of Sidon in additive number theory, and on some related problems. *J. London Math. Soc.* 16 (1941), no. 4, 212–215.
- [6] W. Feller, *An Introduction to Probability Theory and Its Applications*, Vol. II, 2nd ed. Wiley, New York, 1971.
- [7] J. Hou and H. Zhao, An improved upper bound for finite Sidon sets via vector-valued smoothing. arXiv:2607.01169 (2026); version 1 titled “Vector-valued smoothing for finite Sidon sets”.
- [8] B. Lindström, An inequality for  $B_2$ -sequences. *J. Combin. Theory* 6 (1969), no. 2, 211–212.
- [9] K. O’Bryant, A complete annotated bibliography of work related to Sidon sequences. *Electron. J. Combin.* (2004), Dynamic Survey DS11.
- [10] K. O’Bryant, On the size of finite Sidon sets. *Ukrainian Math. J.* 76 (2025), 1352–1368. arXiv:2207.07800.

## Appendix A. Certificate data

The tables record, for each resolution, the number of covering inequalities, the exactly evaluated constants and the two mixing weights, for the original symmetric scan and for the two-sided scan. Every row was verified in rational arithmetic as described in Section 4.1.

| <b>m</b> | <b>L</b> | <b>K</b> | <b>inequalities</b> | <b>a</b>     | <b>b</b>     | <b><math>\gamma^2</math></b> | <b><math>\gamma</math></b> | <b><math>\lambda_1</math></b> | <b><math>\lambda_2</math></b> |
|----------|----------|----------|---------------------|--------------|--------------|------------------------------|----------------------------|-------------------------------|-------------------------------|
| 32       | 4        | 2        | 129                 | 1.294133696  | 0.6874915809 | 0.8897060205                 | 0.943242229                | 0.3592                        | 0.6408                        |
| 64       | 4        | 2        | 257                 | 1.313278418  | 0.6770076151 | 0.8890994897                 | 0.94292072                 | 0.3440                        | 0.6560                        |
| 128      | 4        | 2        | 513                 | 1.3216771522 | 0.6725867335 | 0.8889425185                 | 0.94283748                 | 0.3414                        | 0.6586                        |
| 256      | 4        | 2        | 1025                | 1.3262775741 | 0.6702236072 | 0.8889025398                 | 0.94281628                 | 0.3397                        | 0.6603                        |
| 512      | 4        | 2        | 2049                | 1.3264253589 | 0.6701415970 | 0.8888928083                 | 0.94281112                 | 0.3396                        | 0.6604                        |

*Table A1. Verified symmetric certificates of the original scan. The number of covering inequalities is  $Lm + 1$ . The  $m = 512$  row is the original certificate; its re-optimized successor is in Table 2. Because the optimum has flat directions,  $a$  and  $b$  are not individually converged.*

| <b>m</b> | <b>L</b> | <b>K</b> | <b>inequalities</b> | <b>a</b>     | <b>b</b>     | <b><math>\gamma^2</math></b> | <b><math>\gamma</math></b> | <b><math>\lambda_1</math></b> | <b><math>\lambda_2</math></b> |
|----------|----------|----------|---------------------|--------------|--------------|------------------------------|----------------------------|-------------------------------|-------------------------------|
| 32       | 4        | 2        | 258                 | 1.3131380851 | 0.6770798880 | 0.8890993876                 | 0.9429206688               | 0.1475                        | 0.8525                        |
| 64       | 4        | 2        | 514                 | 1.3230863497 | 0.6718702582 | 0.8889423673                 | 0.9428374024               | 0.1488                        | 0.8512                        |
| 128      | 4        | 2        | 1026                | 1.3281707583 | 0.6692681652 | 0.8889024065                 | 0.9428162103               | 0.1488                        | 0.8512                        |
| 256      | 4        | 2        | 2050                | 1.3305653396 | 0.6680561258 | 0.8888923259                 | 0.9428108643               | 0.1492                        | 0.8508                        |

*Table A2. Verified two-sided certificates; the number of covering inequalities is  $2(Lm + 1)$ .*

**Code and data.** The optimizer, the exact-rational verifier, the resolution scan and the limiting certificate in plain text are available from the author, and will be deposited with the published version.